

Verification 10_Linear_Slip_Weakening_1D_Pore_Pressure

Barnaby Fryer

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator 10_Linear_Slip_Weakening_1D_Pore_Pressure correctly reproduces the results of Garagash and Germanovich (2012) for limiting cases. This is done by digitizing the data of Figure 9 in Garagash and Germanovich (2012) and comparing them to the simulation results produced by the MATLAB and Python simulators in a limiting case (very small production duration) and by checking the boundary between stable and unstable behaviour in another limiting case (very large production duration).

2 Results

2.1 No-production limit

In the case the production duration is set to a small value, Table 1, the simulators should return the injection-only solutions from Garagash and Germanovich (2012). This is shown for both the MATLAB and Python simulators, Figure 1 and 2.

Parameter	Symbol	Value
Number points in crack	N	101
Production pressure	$\Delta\tilde{P}_p$	-0.7
Production duration	$\tilde{t}_p$	0.000001
Injection pressure	$\Delta\tilde{P}_i$	0.75
Residual friction	$\tilde{f}_r$	0.6

Table 1: Simulation parameters for the no-production limit.

2.2 Large-production-duration limit

In the case the production duration is very large, the effective background stress becomes $\frac{\tilde{\tau}_b}{(1-\Delta\tilde{P}_p)}$ as discussed by Fryer et al. (2023). As the boundary between dynamic and stable and unstable behaviour is $\tilde{f}_r$, a way to verify the simulators is to check if the early time behaviour (before the pore pressure changes due to production are undone) has the same boundary between unstable and stable behaviour at $\frac{\tilde{\tau}_b}{(1-\Delta\tilde{P}_p)} = \tilde{f}_r$. This check, Table 2, has been performed here, Figure 3 using a production pressure of $\Delta\tilde{P}_p = -0.2$, meaning the background stress boundary should be at $\tilde{\tau}_b = \tilde{f}_r (1 - \Delta\tilde{P}_p) = 0.72$. This indeed is approximately the case, Figure 3.

3 Conclusion

The simulators have both been verified for large and small production duration limits.

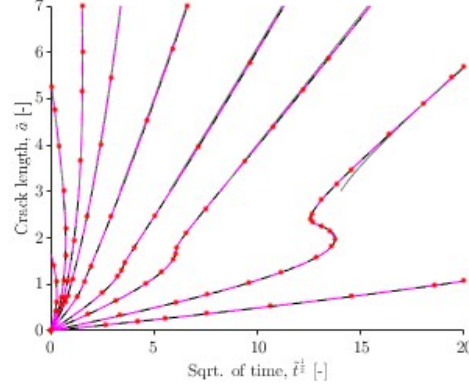


Figure 1: Comparison between MATLAB (black), Python (dashed pink), and small-scale yielding (grey) data with Figure 9 crack-length data of Garagash and Germanovich (2012) (red dots). Each line corresponds to a different background stress, as in Garagash and Germanovich (2012). Note that the production duration was set to a limitingly-small value (0.000001) to reproduce the injection-only simulations in Garagash and Germanovich (2012).

Parameter	Symbol	Value
Number points in crack	N	301
Production pressure	$\Delta \tilde{P}_p$	-0.2
Production duration	$\tilde{t}_p$	1e10
Injection pressure	$\Delta \tilde{P}_i$	0.7
Residual friction	$\tilde{f}_r$	0.6

Table 2: Simulation parameters for the large-production-duration limit.

References

- B. Fryer, M. Lebihain, and M. Violay. Single-well pore pressure preconditioning for Enhanced Geothermal System stimulation. *Journal of Geophysical Research: Solid Earth*, 128, 2023. <https://doi.org/10.1029/2022JB025443>.
- D. Garagash and L. Germanovich. Nucleation and arrest of dynamic slip on a pressurized fault. *Journal of Geophysical Research*, 117, 2012. doi: <https://doi.org/10.1029/2012JB009209>.

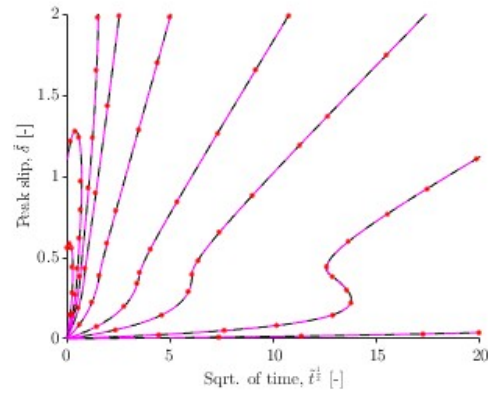


Figure 2: Comparison between MATLAB (black), Python (dashed pink), and small-scale yielding (grey) data with Figure 9 slip data of Garagash and Germanovich (2012) (red dots). Each line corresponds to a different background stress, as in Garagash and Germanovich (2012). Note that the production duration was set to a limitingly-small value (0.000001) to reproduce the injection-only simulations in Garagash and Germanovich (2012).

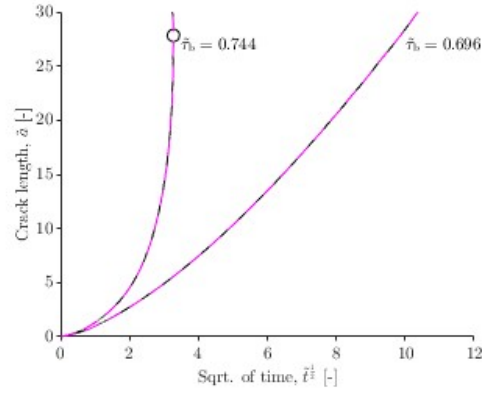


Figure 3: Crack length as function of time for MATLAB (black) and Python (dashed pink). Each line corresponds to a different background stress, as indicated. The white dot denotes a dynamic event. Note that the production duration was set to a limitingly-large value (zero) altering the background stress to a new effective background stress as discussed by Fryer et al. (2023). The pressure change due to production is $\Delta \tilde{P}_p = -0.2$. The boundary of the regime containing early dynamic events and no early dynamic events occurs at an effective background stress of $\tilde{f}_r$, which is 0.6 in these simulations. This corresponds to an actual background stress of $0.6 \left(1 - \Delta \tilde{P}_p\right) = 0.72$. It can be seen in this figure that this is the approximate boundary seen in the simulations as well.